

World Models for LLM Agents

RWML — Teaching AI to Understand Environment Dynamics

From RLHF → to understanding reality itself

RLHF teaches:
“What humans like.”

World Models
teach:
“What reality does.”

Previously... RLHF Taught LLMs Human Preference

We demonstrated that LLMs can be aligned using:

- Preference Learning
- Reward Models
- PPO for Alignment

BUT

Humans and animals still possess something many LLMs lack:



Internal World Models

Your dog may learn from reward signals after only a few experiences... ...but LLM agents still struggle to truly understand:

- consequences
- environment dynamics
- causal changes after actions

LLMs Can Talk... But Can They Predict Consequences?

Good agents do not just generate actions.
They anticipate what happens next.

● LLMs Are Already Good At

- Question Answering
- Code Generation
- Multi-Step Reasoning
- Conversational Fluency

⚠ BUT...

LLMs still struggle in many **agentic environments** that require:

- anticipating action consequences
- adapting to environment dynamics
- understanding hidden causal structure
- long-horizon interaction

Agentic intelligence must consist of more than: language understanding. It also requires: world understanding

Your Dog Has a World Model Too 🐶

🧠 **Biological intelligence has a superpower:** Imagine future consequences BEFORE acting

Intelligence is not just reacting. It is predicting.

Humans and animals learn efficiently because they:

- simulate outcomes
- anticipate consequences
- mentally model environments



Good agents do not only act.

What is a World Model?

They internally simulate outcomes.

(state, action) $\rightarrow$ next state

In short, a world model allows an agent to: “see” what the next state might look like before actually acting.

Imagine your robot plans to: "go to countertop" 

The robot may internally imagine: “There might be a knife there.” *before even moving.*



Beginner

Predict what happens next



Engineer

Learn transition dynamics



Researcher

Approximate the environment transition function

What Previous Methods Fails

Supervised Fine-Tuning (SFT) usually requires:

- expert demonstrations
- curated datasets
- exact target outputs

And the model learns:

“Repeat the expert wording.”

BUT...

Does memorizing exact words mean:

understanding the world?

Probably not.

Human says: “There may be a knife on the countertop.”

SFT learns: repeat this sentence

BUT world modeling should learn: why the knife is likely there

SFT teaches agents to SOUND intelligent.

World models teach agents to THINK ahead.

$$(reasoning, predicted\ next\ state) \sim \pi_{\theta}(s_{\leq t}, a_t)$$

Can We Teach Agents to Understand “Le Monde”?

Can we teach agents through:

- interaction
- consequences
- reinforcement signals

to actually understand:

how the world changes?

YES.

This is the core idea
behind:

**Reinforcement
World Model
Learning (RWML)**

*RWML does not just teach agents to imitate actions.
It teaches agents to anticipate consequences.*

The idea is surprisingly
simple:

👉 Ask the agent to:

- predict the next state
- reason about consequences
- learn from interaction feedback

The Big Shift

RLHF “What humans prefer”	RWML “What reality does”
--	------------------------------------

From preference modeling → to
consequence modeling

RWML Pipeline Flow Diagram

No expert label needed!

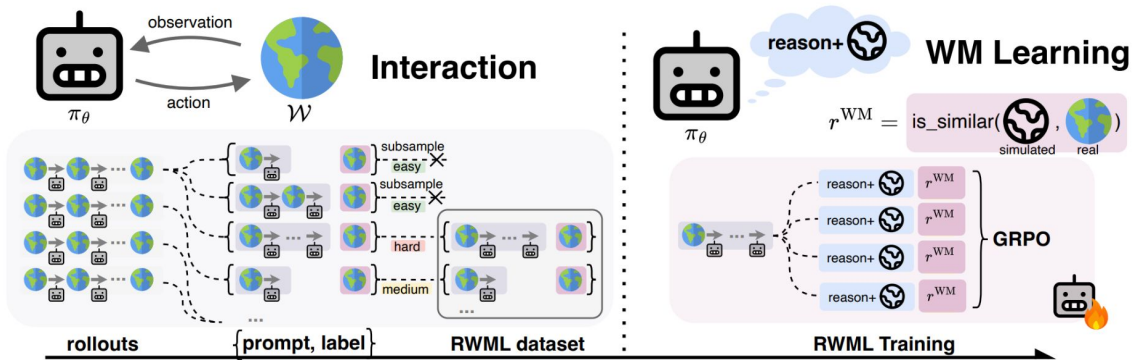


Figure 2. Overview of RWML. Given a target model π_θ , we first collect training data for RWML by using π_θ to gather rollouts $(s_0, a_0, s_1, a_1, \dots, s_T)$ with the environment, and then convert these rollouts into $\langle s_{\leq t}, a_t, s_{t+1} \rangle$ triplets for all t , after subsampling “too easy” samples defined in Equation (1). We then train π_θ to reason as a world model via GRPO, using lightweight reward functions (e.g., embedding-based cosine similarity) to compare the predicted $\hat{s}_{t+1}$ with the real s_{t+1} .

- **rollout**
- **collect transitions**
- **predict next state**
- **compare with reality**
- **RL update**

The Environment Becomes the Teacher

Step 1 — Let The Agent Explore

Collect interaction trajectories:

$(s_0, a_0, s_1, a_1, \dots)$

Example: Kitchen

- go countertop
- see knife
- open drawer
- see spoon

Step 2 — Break Into Transitions

Convert trajectories into:

$(s \leq t, a_t, s_{t+1})$

Meaning:

- current history
- + action
- next state

No experts.
No labels.
No stronger LLMs.

The environment itself
generates the learning
signal.

Meaning > Exact Words

sim(predicted state, real state)

RWML does NOT ask:

“Did you reproduce the exact sentence?”

Instead, it asks:

“Did you understand what happened?”

World models care about understanding reality —
not memorizing exact sentences.

Real state:

“You see a knife on the countertop.”

Prediction 1: -> High Reward

“There is a knife near the counter.”

Prediction 2: -> Low Reward

“The kitchen explodes.”

Why RL Instead of SFT?

SFT: memorizing exam answers

RL: actually understanding the material

 Beginner SFT says: “Repeat exactly what the teacher said.” RL says: “Figure out what actually works.”	 Engineer RL can optimize: • semantic correctness • consequences • behavior quality instead of: exact token matching	 Researcher SFT may cause: token-level collapse RL allows: multiple valid reasoning paths with the SAME semantic outcome.
---	--	---

RL teaches agents to optimize meaning — not just imitate wording.

PPO Again... Except We Call It GRPO

No Q-network.

No V-network.

*Instead:
relative
advantage
within a group*

Traditional Actor-Critic:

- Actor network
- Critic network
- Value estimation
- Q functions

GRPO replaces critics with comparisons.

Actor-Critic: teacher grades your homework

GRPO: students compare answers with each other

GRPO Philosophy

What if we simply:

sample multiple answers

and compare them:
against EACH OTHER?

Example

Prompt: “What happens if the robot opens the drawer?”

Model samples:

- “find spoon”
- “find knife”
- “kitchen explodes” 😭

Then GRPO says: “Which prediction makes more sense?”

Everything Looks Fine... Until The Agent Thinks The Exam Is Too Easy

Many transitions are:

- repetitive
- trivial
- easy to predict

Example:

"go to kitchen" → "arrive at kitchen"

Your agent quickly learns: "I already know this 😴"

Training on easy transitions repeatedly:

- wastes compute
- teaches little new information
- weakens learning signal

 **RWML Solution**

Remove: easy questions

Focus on: surprising / difficult transitions

Intelligence grows faster when learning focuses on difficult situations.

ALFWorld Example (Kitchen)

(1) ALFWorld

= Welcome to TextWorld, ALFRED! =

You are in the middle of a room. Looking quickly around you, you see a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, (...omitted), a stoveburner 2, a stoveburner 1, and a toaster 1.

Your task is to: put a knife in sidetable.

(1a) Before WMRL



<think>To complete the task of putting a knife in the sidetable, I need to find a knife first. The most logical places to find a knife would be in the drawers, cabinets, or countertop. I will start with the drawers.</think>
go to drawer 1

The drawer 1 is closed.



...(a few turns omitted)



<think>So far, I've gathered that: **Drawer 1 is now open but contains nothing; Drawer 2 also does not contain a knife...**</think>

examine cabinet 3

max 30 steps reached



(1b) After WMRL



<think>The task is to put a knife in the sidetable. First, I need to find a knife. **It is likely to be on the countertop 1.** After finding the knife, I need to go to the sidetable 1.</think>

go to countertop 1

You arrive at countertop 1. On the countertop 1, you see a apple 2, a bread 2, (...omitted), **a knife 4, a knife 1**, a potato 1, a saltshaker 1, a spatula 1, a spoon 2, and a spoon 1.



<think>The task is to put a knife in the sidetable. First, I need to pick up the knife 1 from the countertop 1. After that, I will go to the sidetable 1 and put the knife on it.</think>

take knife 1 from countertop 1

You pick up the knife 1 from the countertop 1.



...(a few turns omitted)

<think>The task is to put a knife in the sidetable 1. The user is currently at the sidetable 1...</think>

move knife 1 to sidetable 1

success in 5 steps



Tau2Bench Example (Customer Agent)

(2) Tau2-Bench

You are a customer service agent... Available functions:
{
 "type": "function",
 "function": {
 "name": "get_customer_by_phone",
 "description": "Finds a customer by their primary contact or ..."
 }
}
(...more tools and instructions omitted)

user: Hi, I'm having an issue with my phone. It's been showing 'No Service' for the past few hours, and I'm not sure what's wrong. I've tried restarting it already, but that didn't help. Could you please assist me in fixing this?

(2a) Before WMRL



<think>Okay, the user is experiencing a 'No Service' issue...First, I need to check if I can identify the customer... So the response should ask the user to provide their phone number... Once I have that, I can proceed to check their line status, bill status, or other relevant details.</think>

Could you please provide your phone number, customer ID, or full name along with your date of birth?

My name is John Smith, and my phone number is 555-123-2002.



...(a few turns omitted)

<think>...The user is confused about two payments they made...

Let me go through the steps again...

max 30 steps reached



(2b) After WMRL



<think>Okay, the user is experiencing 'No Service'... First, according to the tech support policy, I need to check for common service issues...

Wait, since the user mentioned they tried restarting, it is possible Airplane Mode is on. If it's on, turning it off might resolve it...</think>

To help resolve the 'No Service' issue, I'll need some details. Could you please provide your phone number or customer ID?

If you're unsure, you can also check your phone's status bar for icons like "✈️ Airplane Mode" or "📶" signal strength, and let me know if those are enabled/disabled.

I checked my status bar, and it shows that Airplane Mode is turned on. I didn't realize that! Could that be why I have no service?



...(a few turns omitted)



<think>Okay, let me go through the user's problem... After finding Airplane Mode on, they turned it off, but the problem persisted. Then they checked the network status and saw the SIM was "locked_pin."

So the user is now facing a SIM lock issue, which is a security feature. According to the policy, I should transfer the user to a human agent with summary of the issue...

success in 4 steps



The Agent Improved... Without Being Told The Correct Answer

ALFWorld

+19.6 improvement

Tau²Bench

+7.9 improvement



And They Did This WITHOUT...

-  Expert demonstrations
-  Stronger teacher LLMs
-  Task-success reward signals
-  Human annotations



What Did The Agent Learn From Then?

Simply:

interaction + consequences

The world itself became the supervisor.



Fun Analogy

Nobody told the agent: “Here is the correct answer.”

The agent basically learned by:
“touching grass in the environment”

Why Does Fine-Tuning Make Models Forget?

Human Intelligence Is Surprisingly Stable

If humans study:

 chess for 10 months

we usually do NOT suddenly forget:

 calculus

Why SFT Causes Forgetting

SFT forces models to: memorize exact token patterns

Large parameter updates may overwrite:

- previous knowledge
- reasoning skills
- general capabilities

But LLMs Often Do...

After aggressive SFT:

- math ability drops
- coding ability drops
- reasoning quality drops

This is called: catastrophic forgetting

***RWML teaches new skills
without erasing old
intelligence as much.***

Why Are World Models Such a Big Deal?

World Models Enable...

 Planning “What should I do next?”	 Anticipation “What might happen if I act?”	 Efficient Exploration “Which actions are worth trying?”	 Long-Horizon Reasoning “How will this affect the future?”
--	---	---	--

Without world models: the agent is basically speedrunning life blindly

With world models: the agent mentally simulates outcomes before acting

Prediction is a core ingredient of intelligence.

Where Is AI Heading Next?



Evolution of AI

Chatbots

→ Agents

→ World Models

→ Autonomous Systems



Future LLM Agents May Need...



Memory



Planning



Simulation



Environment Understanding



Long-Term Decision Making



Key Insight

Future AI systems may not just:

generate fluent language

They may need to:

understand consequences
in dynamic environments

The future of AI is not just conversation.

It is interaction with reality.

Final Takeaways

- RLHF teaches: human preference
- RWML teaches: environment dynamics
- World models improve: planning and decision-making
- Self-supervised RL can teach: powerful agent behavior without experts

The BIG Philosophical Shift

From predicting tokens
→ to predicting consequences

FINAL LINE (VERY BIG FONT)

The future of AI is not just
generating text —

it is understanding reality.